Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 27: Approved Solutions

Proof by Contradiction

Covers Chartrand & Zhang, Section 3.7

Status of this set

Ungraded practice. Solutions released with it. Write every proof in complete sentences, and begin each with "Assume, to the contrary, that…"

Part I: How to Practice This Material

The template

1. **Assume the negation** of what you want to prove. For an implication, assume the hypothesis holds *and* the conclusion fails.
2. **Name** anything whose existence you have just assumed.
3. **Derive** a statement that contradicts a known fact or one of your assumptions.
4. **Conclude** that the assumption was false, so the original statement is true.

When to prefer it

Reach for contradiction when there is **no implication to reverse** — "$\sqrt{2}$ is irrational," "there is no largest…" If the statement is an implication and the contrapositive works with the same algebra, the contrapositive is cleaner.

Three mistakes to expect

1. **Negating incorrectly.** The negation of $P \Rightarrow Q$ is $P \wedge (\sim Q)$, not $(\sim P) \Rightarrow (\sim Q)$.
2. **Never reaching a contradiction.** Say explicitly what is contradicted.
3. **Using contradiction where contrapositive is simpler.** Legal, but longer.

Part II: Practice Problems

Section A — "There is no…"

1. Prove that there is no largest even integer.

Approved solution

Proof. Assume, to the contrary, that there is a largest even integer; call it n. Then $n = 2k$ for some integer k. Consider $m = n + 2 = 2(k + 1)$. Since $k + 1$ is an integer, m is even, and $m > n$. This contradicts that n is the largest even integer. ■

The same shape as "no smallest positive real": assume the extreme object exists, name it, and build a more extreme one from it.

2. Prove that there is no largest rational number less than 1.

Approved solution

Proof. Assume, to the contrary, that there is a largest rational number r with $r < 1$. Consider

$$s = \frac{r+1}{2}.$$

Since r is rational, s is rational (a sum and quotient of rationals, with nonzero denominator). Since $r < 1$, we have $r < s < 1$: it lies strictly between r and 1.
So s is a rational number less than 1 and greater than r — contradicting that r was the largest. ■

Taking the midpoint is the standard way to squeeze a new element between two others. It fails in $\mathbb{Z}$ — there *is* a largest integer less than 1, namely 0 — because the midpoint of two integers need not be an integer.

Section B — Irrationality

3. You may use the fact: *for an integer n, if n^2 is divisible by 5, then n is divisible by 5.* Prove that $\sqrt{5}$ is irrational.

Approved solution

Proof. Assume, to the contrary, that $\sqrt{5}$ is rational. Then $\sqrt{5} = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$, and a/b in lowest terms.
Squaring, $5 = a^2/b^2$, so $a^2 = 5b^2$. Hence a^2 is divisible by 5, and by the given fact, a is divisible by 5. Write $a = 5c$ for some integer c.
Then $25c^2 = 5b^2$, so $b^2 = 5c^2$. Hence b^2 is divisible by 5, so b is divisible by 5.
So a and b share the factor 5, contradicting that a/b is in lowest terms. ■

The same proof as $\sqrt{2}$, with 5 in place of 2. The only change is the fact used: "$5 \mid n^2 \Rightarrow 5 \mid n$" instead of Theorem 3.24.

4. Prove: if a is rational and b is irrational, then $a + b$ is irrational.

Approved solution

Proof. Assume, to the contrary, that a is rational, b is irrational, and $a + b$ is rational. Then $b = (a + b) - a$ is a difference of two rational numbers.
Write $a + b = p/q$ and $a = r/s$ with $p, q, r, s \in \mathbb{Z}$ and $q, s \neq 0$. Then

$$b = \frac{p}{q} - \frac{r}{s} = \frac{ps - qr}{qs},$$

where $ps - qr$ and qs are integers and $qs \neq 0$. So b is rational — contradicting that b is irrational. ■

The negation of the implication is "a rational, b irrational, and $a + b$ rational." Getting that negation right is the whole first step.

Section C — Parity

5. Prove by contradiction: let $n \in \mathbb{Z}$; if $3n - 11$ is odd, then n is even.

Approved solution

Proof. Assume, to the contrary, that $3n - 11$ is odd and n is odd. Then $n = 2k + 1$ for some integer k, and

$$3n - 11 = 6k + 3 - 11 = 6k - 8 = 2(3k - 4).$$

Since $3k - 4$ is an integer, $3n - 11$ is even — contradicting that it is odd. ■

Now compare. A proof by contrapositive would assume n odd and show $3n - 11$ even — identical algebra, one fewer assumption. Here both methods are legal and the contrapositive is slightly cleaner. Contradiction is the better tool only when there is no implication to reverse.

Section D — Writing

6. In five to eight complete sentences, explain why deriving a contradiction proves the original statement. Name the rule of inference from Week 3 that makes this legitimate.

Approved solution

Model answer. In a proof by contradiction we assume $\sim R$ and derive a contradiction C, a statement of the form $P \wedge (\sim P)$ that is false in every row. That establishes the implication $(\sim R) \Rightarrow C$. Since C is false, and an implication with a false conclusion is true only when its hypothesis is also false, $\sim R$ must be false. This inference is modus tollens: from $(\sim R) \Rightarrow C$ and $\sim C$, we conclude $\sim(\sim R)$. Applying double negation, that is R. So proof by contradiction is not a new logical principle but a combination of modus tollens and double negation, both of which were established in the first three weeks.

Full credit requires naming modus tollens and explaining why a false conclusion forces a false hypothesis.

Section E — Challenge (optional)

7. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Approved solution

Proof. Assume, to the contrary, that $\sqrt{2} + \sqrt{3} = r$ is rational. Squaring,

$$r^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}, \qquad \text{so} \qquad \sqrt{6} = \frac{r^2 - 5}{2}.$$

The right side is rational, since r is. So $\sqrt{6}$ would be rational.

But $\sqrt{6}$ is irrational (Chartrand Section 3.7 Exercise 13). This is a contradiction. ■

The strategy is worth naming: reduce to a case already known. We did not prove $\sqrt{2}+\sqrt{3}$ irrational from scratch; we showed its rationality would force $\sqrt{6}$ to be rational, which it is not. Reusing earlier results this way is how longer proofs stay short.

Instructor notes

- **Problem 5 compares methods directly.** Same algebra as the contrapositive; one extra assumption. Students should leave knowing when each is preferable.
- **Problem 3** is $\sqrt{2}$ with 5 substituted, and makes clear the proof depends on one divisibility fact.
- **Problem 2's closing remark** explains why the midpoint argument fails in $\mathbb{Z}$. Good for strong students.
- **Problem 6** is the logical justification — modus tollens from Lecture 7.
- **Problem 7** reduces to a known case, a strategy worth naming.
- Reserved for Homework 10: Section 3.7 Exercises 2, 4, 6, 8 (even).